

Phase 5 — Sustainability & Analytics Spec

PREPARED BY Energy Hippo implementation team

GHG emissions, ENERGY STAR ratings, weather regression, and benchmarks — the reporting that needs config + a processing/calc result rather than raw load. Lights up GHG History/Variance/Audit (3), ES Rating Comparison/History/RequestLog (3), WeatherRegression + RegressionSummary, and BenchmarkRanking/MonthlyEUI/ PerformanceSnapshot.

Entry path (see [entry-path-and-gaps.pdf](#)): GHG sources/factors/meter-assignment/company-config and ENERGY STAR property setup + rating request are **UI-DIRECT** (ES is fully local — no PM account needed to set up). Weather regression is **UI-THEN-RUN**. **Critical gap:** ENERGY STAR **scores** (ESResponseFacility) are **EXTERNAL** — they only arrive from the Portfolio Manager API via the sync task; they **cannot be hand-entered** (B1), so score-dependent reports need a sandbox/real PM account or stay empty. **Benchmark definitions** have no create UI (A6). GHG company config lacks Scope 1/2/3 toggles (content gap).

Source-model extension

Add: `emission_factor_source` (ref to EEM source), `emission_factor` (source, measure_type, emission_type, value, unit, effective_date), `emission_meter_map` (meter, category, start/end), `emission_company_config` (company, scopes, year); `es_rating_response` (es_property, period, score, eligible, metrics); `weather_regression_result` (meter/point, period, weather_pt, intercept, hdd_factor, cdd_factor, r2, cvrmse); `benchmark_definition` (company, name, formula, unit).

1. GHG / emissions

EEMSuite (verified): EmissionType (18; CO₂=GWP 1, CH₄=28, N₂O=265...), EmissionCategory (1 Indirect, 2 Stationary, 3 Mobile, 4 Fugitive, 5 Process, 6 De Minimis), EmissionFactorSource (incl. **4 eGRID**, **3 Non-electric commodities**, **10 Steam Plant**, 1 US EIA). GHG = usage × factor × GWP.

Factor sources → Northlake commodities:

commodity	EmissionFactorSource	category (scope)	factor (reuse provider values)
electric (VED)	4 eGRID (CAMX)	1 Indirect (Scope 2)	0.231 kgCO ₂ e/kWh (+ CH ₄ , N ₂ O)
natural gas (SGU)	3 Non-electric commodities	2 Stationary (Scope 1)	5.31 kgCO ₂ e/therm
chilled water / steam (NTP)	10 Steam Plant	2 Stationary (Scope 1, plant)	from plant gas/electric input factors
water/sewer (RCU)	3 Non-electric	1 Indirect (Scope 3)	3.2 kWh/kgal embedded → grid factor
solar (HOS)	—	Scope 2 reduction	avoided 0.231 kgCO ₂ e/kWh (market-based, RECs retained)

Add `emission_factor` rows (with `EmissionFactorHistory` effective dates for a 2024 → 2025 factor change), `emission_meter_map` (each meter → its category), and `emission_company_config` (Scopes 1+2+3 enabled, reporting years 2024–2025). GHG reports then compute from the existing usage.

2. ENERGY STAR ratings

Property mapping already exists (13 properties, exact `PrimaryFunction`). Add **rating requests + responses (scores)** — normally from the PM API; seed `es_rating_response` directly:

- **Quarterly** ENERGY STAR 1–100 scores 2024–2025 for **eligible** properties (e.g. 72 → 78 trend showing improvement). In the US a score exists only for Office, Residence Hall/Dormitory and Multifamily Housing (Library is score-eligible only in Canada); College/University, Laboratory, Food Service, Fitness Center/Health Club/Gym, Parking, Other - Utility and the other types in the mapping are metrics only. Per energystar.gov a property is eligible only when more than 50 percent of its gross floor area (excluding parking) is an eligible type and the combined floor area of uses without a score does not exceed 25 percent. Lakeview Residence Hall and Cedar Row Apartments are expected to score. Admin Hall is Office-primary, but its classroom wing is 29 percent of its floor area, so it gets metrics only.
- **Ineligible** properties get **metrics but no 1–100 score** (`eligible=false`) — realistic, and exercises the no-score path.
- A few requests with error/pending status for `rp_ESRatingRequestLog`.

3. Weather regression / normalization

Produce `weather_regression_result` (→ `FF_wnRegressionBestResults`) for the metered buildings from existing interval + HDD/CDD (KSAC), model `Use = intercept + HDD·f_h + CDD·f_c`:

meter/point	weather inputs	expect
Science Center electric kWh	HDD + CDD	cooling-driven, $R^2 \geq 0.8$
Student Center demand kW	CDD	event/occupancy noise, $R^2 \sim 0.6$
Admin Hall gas therms	HDD	heating-driven, $R^2 \geq 0.85$
NTP chilled-water ton-hours	CDD	strong cooling fit

This also fills the 3 baseline **shells** (Phase 0/1) with computed values and enables the weather-normalized variants of `ScatterPlot/UsageVariance`.

4. Benchmarks / EUI

`benchmark_definition` rows (floor areas already exist):

benchmark	formula	unit
Site EUI	annual kBtu ÷ gross floor area	kBtu/sqft
Cost intensity	annual cost ÷ gross floor area	\$/sqft
GHG intensity	annual kgCO ₂ e ÷ gross floor area	kgCO ₂ e/sqft

Peer set = the campus and Cedar Row buildings + Town Center buildings, so `rp_BenchmarkRanking` ranks meaningfully and `rp_PerformanceSnapshot` / `rp_MonthlyEUI` populate. Labs (high EUI) vs library/office (low) gives spread.

expectedAssertions

- `ghg_computed` — GHG History = $\Sigma(\text{usage} \times \text{factor} \times \text{GWP})$ per scope; electric = Scope 2, gas/steam = Scope 1, water = Scope 3; solar shows avoided emissions.
- `ghg_factor_change` — a 2025 factor differs from 2024 (GHGVariance non-zero).
- `es_scores_present` — eligible properties have quarterly 1–100 scores (RatingHistory trend); ineligible properties show metrics with no score; RequestLog shows a pending/error request.
- `regression_fit` — each `weather_regression_result` has R^2 and CV-RMSE in plausible ranges; baselines now have computed values.
- `weather_normalized_use` — UsageVariance weather-normalized column differs from raw and is flatter year-over-year.
- `benchmark_ranking` — buildings rank by EUI with labs highest; $\text{EUI} = \text{use} \div \text{area}$ matches Phase 1 `rp_MonthlyEUI`.